LUCAS NOIROT (PETIT)

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PROFILE

Passionate about Computer Science and Mathematics, I specialize in bridging the gap between abstract concepts and concrete, practical solutions.

Personal Attributes

Rigorous · Abstract Thinking · Self-learner · Proactive · Team Player · Creative

Interests

Origami · Board Games · Astronomy

TECHNICAL SKILLS

Formal Methods & Cryptography

Formal Methods (Coq) · Security Proofs

Optimization & Modeling

Mixed-Integer Programming (MIP) · Constraint Programming · Heuristics · Complexity Analysis

Development

C/C++ · Rust · OCaml · Python · SageMath · Java · HTML/CSS/JavaScript

Tools & Environment

Linux · Git · Docker · LaTeX

LANGUAGES

- **French** (Native)
- **English** (C1 – Full Professional Proficiency)

R&D ENGINEER - ALGORITHMS & APPLIED MATHEMATICS

EDUCATION

2025 — **M.SC. IN COMPUTER SCIENCE – ALGORITHMS**
University of Montpellier, France
Cryptography – Computer Algebra – Optimization

2023 — **M.SC. IN MATHEMATICS (1ST YEAR)**
University of Burgundy, France
Linear Algebra – Field Theory – Probability Theory

2022 – 2019 — **B.SC. IN MATHEMATICS**
University of Burgundy, France
Ranked 5th out of 55 (Top 10%)

EXPERIENCE

2025 *(5 months)* — **RESEARCH INTERNSHIP IN CRYPTOGRAPHY**
LIRMM Laboratory – ECO Team

Topic: Lattice Reduction – Adapting strategies from the polynomial case. Developed reduction algorithms using **SageMath**. Conducted complexity analysis and benchmarked performance between polynomial and integer arithmetic.
enpeluche.github.io/lattice-reduction-internship/

2022 — **PRIVATE TUTOR**
Dijon, France – Freelance
Provided Mathematics and Physics tutoring to middle and high school students.

PERSONAL PROJECTS

Workforce Scheduling & Routing (WSRP)

Designed a hybrid solver for technician routing under skill constraints. Implemented a **heuristic** in **Java** coupled with Constraint Programming (**OR-Tools**).
enpeluche/wsrp-lns-solver

Coq-Verified Red-Black Tree

Formalized and proved balancing invariants in **Coq**, with automatic extraction to an efficient, certified **OCaml** module.
enpeluche/coq-verified-rbt

Constraint Solving Engine Architecture (CSP)

Developed a propagation engine from scratch in **C++**. Implemented Arc-Consistency algorithms (**AC-3**).
enpeluche/constraint-puzzle-solver

Zero-Knowledge Proof (ZKP) Protocol

Implemented the **Schnorr authentication protocol** over Elliptic Curves. Prototyped algebraically in **SageMath** and developed a high-performance version in **C++**.
enpeluche/zkp-schnorr-cpp

Post-Quantum Messaging (Lattice-based)

Implemented a quantum-resistant encrypted chat application. Developed a **C++** key exchange protocol based on the **LWE (Learning With Errors)** problem over lattices. Built an asynchronous Client/Server architecture.

https://github.com/enpeluche/